

UNIT -II

FORE CASTING

INTRODUCTION

The growing competition, frequent changes in customer's demand and the trend towards automation demand that decisions in business should not be based purely on guesses rather on a careful analysis of data concerning the future course of events. More time and attention should be given to the future than to the past, and the question 'what is likely to happen?' should take precedence over 'what has happened?' though no attempt to answer the first can be made without the facts and figures being available to answer the second. When estimates of future conditions are made on a systematic basis, the process is called forecasting and the figure or statement thus obtained is defined as forecast.

In a world where future is not known with certainty, virtually every business and economic decision rests upon a forecast of future conditions. Forecasting aims at reducing the area of uncertainty that surrounds management decision-making with respect to costs, profit, sales, production, pricing, capital investment, and so forth. If the future were known with certainty, forecasting would be unnecessary. But uncertainty does exist, future outcomes are rarely assured and, therefore, organized system of forecasting is necessary. The following are the main functions of forecasting:

- The creation of plans of action.
- The general use of forecasting is to be found in monitoring the continuing progress of plans based on forecasts.
- The forecast provides a warning system of the critical factors to be monitored regularly because they might drastically affect the performance of the plan.

It is important to note that the objective of business forecasting is not to determine a curve or series of figures that will tell exactly what will happen, say, a year in advance, but it is to make analysis based on definite statistical data, which will enable an executive to take advantage of future conditions to a greater extent than he could do without them. In forecasting one should note that it is impossible to forecast the future precisely and there always must be some range of error allowed for in the **forecast**.

FORECASTING FUNDAMENTALS

Forecast: A prediction, projection, or estimate of some future activity, event, or occurrence.

Types of Forecasts

- Economic forecasts
 - Predict a variety of economic indicators, like money supply, inflation rates, interest rates, etc.
- Technological forecasts
 - Predict rates of technological progress and innovation.
- Demand forecasts
 - Predict the future demand for a company's products or services.

Since virtually all the operations management decisions (in both the strategic category and the tactical category) require as input a good estimate of future demand, this is the type of

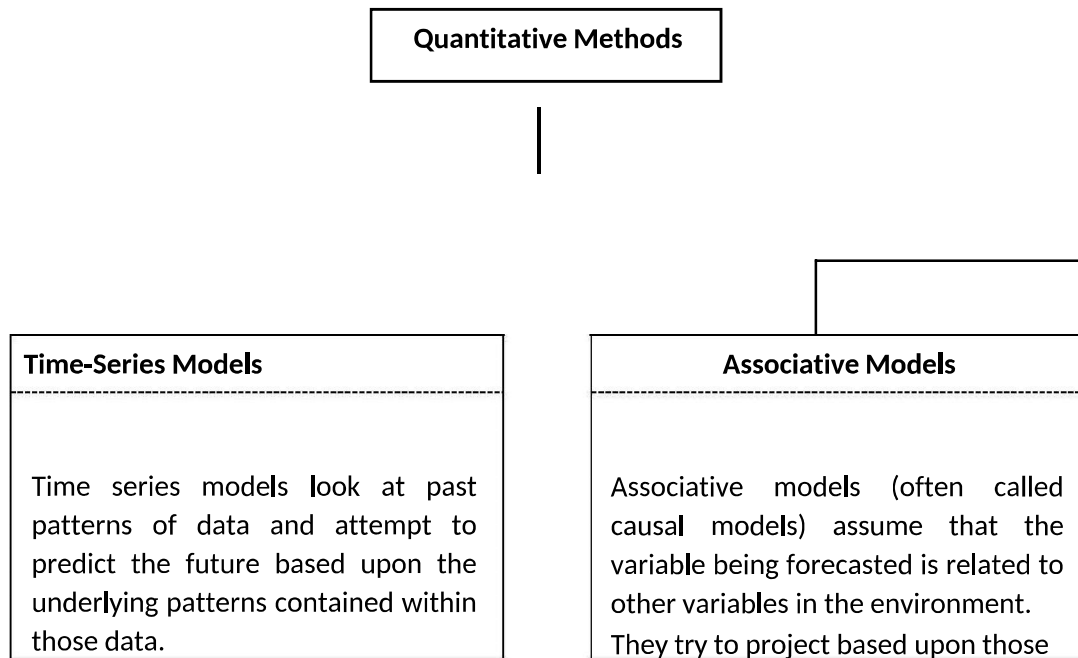
forecasting that is emphasized in our textbook and in this course.

TYPES OF FORECASTING METHODS

Qualitative methods: These types of forecasting methods are based on judgments, opinions, intuition, emotions, or personal experiences and are subjective in nature. They do not rely on any rigorous mathematical computations

Quantitative methods: These types of forecasting methods are based on mathematical (quantitative) models, and are objective in nature. They rely heavily on mathematical computations.

QUANTITATIVE FORECASTING METHODS



TIME SERIES MODELS

Model	Description
Naïve	Uses last period's actual value as a forecast
Simple Mean (Average)	Uses an average of all past data as a forecast
Simple Moving Average	Uses an average of a specified number of the most recent observations, with each observation receiving the same emphasis (weight)
Weighted Moving Average	Uses an average of a specified number of the most recent observations, with each observation receiving a different emphasis (weight)
Exponential Smoothing	A weighted average procedure with weights declining exponentially as data become older
Trend Projection	Technique that uses the least squares method to fit a straight line to the data
Seasonal Indexes	A mechanism for adjusting the forecast to accommodate any seasonal patterns inherent in the data

DECOMPOSITION OF A TIME SERIES

Patterns that may be present in a time series

Trend: Data exhibit a steady growth or decline over time.

Seasonality: Data exhibit upward and downward swings in a short to intermediate time frame (most notably during a year).

Cycles: Data exhibit upward and downward swings in over a very long time frame.

Random variations: Erratic and unpredictable variation in the data over time with no discernable pattern.

ILLUSTRATION OF TIME SERIES DECOMPOSITION

Hypothetical Pattern of Historical Demand

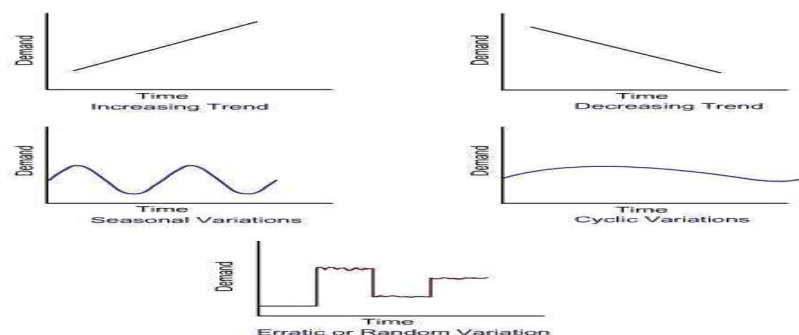
Dependent versus Independent Demand

Demand of an item is termed as independent when it remains unaffected by the demand for any other item. On the other hand, when the demand of one item is linked to the demand for another item, demand is termed as dependent. It is important to mention that only independent demand needs forecasting. Dependent demand can be derived from the demand of independent item to which it is linked.

Business Time Series

The first step in making a forecast consists of gathering information from the past. One should collect statistical data recorded at successive intervals of time. Such a data is usually referred to as time series. Analysts plot demand data on a time scale, study the plot and look for consistent shapes and patterns. A time series of demand may have constant, trend, or seasonal pattern ([Figure 1](#))

Figure 1: Business Time Series



or some combination of these patterns. The forecaster tries to understand the reasons for such changes, such as,

- Changes that have occurred as a result of general tendency of the data to increase or decrease, known as secular movements.
- Changes that have taken place during a period of 12 months as a result in changes in climate, weather conditions, festivals etc. are called as seasonal changes.
- Changes that have taken place as a result of booms and depressions are called as cyclical variations.
- Changes that have taken place as a result of such forces that could not be predicted (like flood, earthquake etc.) are called as irregular or erratic variations.

Quantitative Approaches of Forecasting

Most of the quantitative techniques calculate demand forecast as an average from the past demand. The following are the important demand forecasting techniques.

- Simple average method: A simple average of demands occurring in all previous time periods is taken as the demand forecast for the next time period in this method. (Example [1](#))

Example 1

Simple Average :

A XYZ television supplier found a demand of 200 sets in July, 225 sets in August & 245 sets in September. Find the demand forecast for the month of October using simple average method.

The average demand for the month of October is

$$\begin{aligned} SA &= \left(\frac{D1 + D2 + D3}{3} \right) \\ &= \left(\frac{200 + 225 + 245}{3} \right) \\ &= 223.33 \\ &\approx 224 \text{ units} \end{aligned}$$

- Simple moving average method: In this method, the average of the demands from several of the most recent periods is taken as the demand forecast for the next time period. The number of past periods to be used in calculations is selected in the beginning and is kept constant (such as 3-period moving average). ([Example 2](#))

Simple Moving Average :

A XYZ refrigerator supplier has experienced the following demand for refrigerator during past five months.

Month	Demand
February	20
March	30
April	40

May	60
June	45

Find out the demand forecast for the month of July using five-period moving average & three-period moving average using simple moving average method.

$$MA_n = \frac{\sum_{i=1}^n D_i}{n}$$

For five period average (i.e. n=5)

$$MA_5 = \frac{20+30+40+60+45}{5}$$

$$= 29 \text{ units}$$

For three period average (i.e. n=3)

$$MA_3 = \frac{40+60+45}{3}$$

$$= 48.33$$

$$\approx 49 \text{ units}$$

- Weighted moving average method: In this method, unequal weights are assigned to the past demand data while calculating simple moving average as the demand forecast for next time period. Usually most recent data is assigned the highest weight factor. ([Example 3](#))

Example 3

Weighted Moving Average Method :

The manager of a restaurant wants to make decision on inventory and overall cost. He wants to forecast demand for some of the items based on weighted moving average method. For the past three months he experienced a demand for pizzas as follows:

Month	Demand
October	400
November	480
December	550

Find the demand for the month of January by assuming suitable weights to demand data.

$$WMA = \sum_{i=1}^n C_i D_i$$

C_i = Weights for Periods

D_i = Demand for Periods

Let $C_1=0.25$, $C_2=0.3$, $C_3=0.5$

$\therefore WMA = C_1 D_1 + C_2 D_2 + C_3 D_3$

$$= 0.25 * 400 + 0.3 * 480 + 0.5 * 550$$

$$= 100 + 144 + 275$$

$$= 519 \text{ units}$$

- Exponential smoothing method: In this method, weights are assigned in exponential order. The weights decrease exponentially from most recent demand data to older demand data. ([Example 4](#))

Example 4

Exponential Smoothing:

One of the two wheeler manufacturing company experienced irregular but usually increasing demand for three products. The demand was found to be 420 bikes for June and 440 bikes for July. They use a forecasting method which takes average of past year to forecast future demand. Using the simple average method demand forecast for June is found as 320 bikes (Use a smoothing coefficient 0.7 to weight the recent demand most heavily) and find the demand forecast for August.

$$F_t = \alpha D_{t-1} + (1 - \alpha) F_{t-1}$$

where α = Smoothing Coefficient

D_{t-1} = Actual Demand for Recent Period

F_{t-1} = Demand Forecast for Recent Period

F_t = Forecast of Next Period Demand

for July:

$$= 0.7(420) + (1 - 0.7)320$$

$$= 294 + 96$$

$$= 390 \text{ units}$$

for August:

$$= 0.7(440) + (1 - 0.7)390$$

$$= 308 + 117$$

$$= 425 \text{ units}$$

- Regression analysis method: In this method, past demand data is used to establish a functional relationship between two variables. One variable is known or assumed to be known; and used to forecast the value of other unknown variable (i.e. demand). ([Example 5](#))

Example 5

Regression Analysis :

Farewell Corporation manufactures Integrated Circuit boards(I.C board) for electronics devices. The planning department knows that the sales of their client goods depends on how much they spend on advertising, on account of which they receive in advance of expenditure. The planning department wish to find out the relationship between their clients advertising and sales, so as to find demand for I.C board.

The money spend by the client on advertising and sales (in dollar) is given for different periods in following table :

Period(t)	Advertising (X _t) \$(1,00,000)	Sales (D _t) \$(1,000.000)	D _t ²	X _t ²	X _t D _t
1	20	6	36	400	120
2	25	8	64	625	200
3	15	7	49	225	105
4	18	7	49	324	126
5	22	8	64	484	176
6	25	9	81	625	225
7	27	10	100	729	270
8	23	7	49	529	161
9	16	6	36	256	96
10	20	8	64	400	120
Σ	211	76	592	4597	1599

$$\begin{aligned}b &= \frac{n(\sum X_t D_t) - (\sum X_t)(\sum D_t)}{n(\sum X_t^2) - (\sum X_t)^2} \\&= \frac{10(1599) - (211)(76)}{10(4597) - (211)^2} \\&= \frac{15990 - 16036}{45970 - 44521} \\&= \frac{-46}{1449} = -0.0317 \\a &= \sum D_t - b \sum X_t \\&= \frac{76 - (-0.0317)211}{10} \\&= 8.268\end{aligned}$$

Relationship between future sales F_t and advertising cost X_t is

$$\begin{aligned}F_t &= a + bX_t \\&= 8.268 - 0.0317X_t\end{aligned}$$

Error in Forecasting

Error in forecasting is nothing but the numeric difference in the forecasted demand and actual demand. [MAD](#) (Mean Absolute Deviation) and [Bias](#) are two measures that are used to assess the accuracy of the forecasted demand. It may be noted that MAD expresses the magnitude but not

the direction of the error.

$$\text{MAD} = \frac{\text{sum of the absolute value of forecast error for all periods}}{\text{number of periods}}$$

$$= \frac{\sum_{i=1}^n |\text{forecast error}_i|}{n}$$

$$= \frac{\sum_{i=1}^n |(\text{forecasted demand} - \text{actual demand})_i|}{n}$$

where n is the number of periods.

$$\text{Bias} = \frac{\text{sum of all forecast errors for all periods}}{\text{number of periods}}$$

$$= \frac{\sum_{i=1}^n (\text{forecast demand} - \text{actual demand})_i}{n}$$

where n is the number of periods.